

Solution Requirements

Date	28 JUNE 2028
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users (Farmers/Admin) can securely log in using username and password to access the application.</i>	High
2	Authorization levels	<i>Farmers can submit soil data and receive crop recommendations. Admin can manage datasets, monitor predictions, and update the ML model.</i>	High
3	External interfaces	<i>The system uses a Crop Recommendation Dataset (CSV), Flask Web Framework, and Machine</i>	High

		<i>Learning libraries such as Scikit-learn, Pandas, and NumPy.</i>	
4	Transactions processing	<i>The application accepts soil and environmental parameters, preprocesses the data, performs prediction using the ML model, and displays the recommended crop.</i>	High
5	Reporting	<i>Generates crop recommendation reports, prediction results, crop suitability information, and farming insights for users.</i>	Medium
6	Business rules, etc.	<i>All input values must be valid. The system predicts crops only for supported soil and environmental conditions available in the trained dataset.</i>	High
7	Compliance to laws or regulations	<i>User information and prediction data are securely stored. The application follows basic data privacy and security practices.</i>	Medium
8	Other	<i>User information and prediction data are securely stored. The application follows basic data privacy and security practices.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>The system should process user input and generate crop recommendations quickly.</i>	<i>Prediction displayed within 2–3 seconds</i>
2	Scalability	<i>The application should support an increasing number of users and future dataset expansion.</i>	<i>Supports 1000+ users without performance degradation</i>
3	Security & Data Privacy	<i>User credentials and prediction data should be protected from unauthorized access.</i>	<i>Password encryption and secure data storage</i>
4	Reliability & Availability	<i>The application should remain available with minimal downtime.</i>	<i>99% system availability</i>
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy for farmers to use on desktop or mobile devices</i>	<i>User-friendly interface with responsive design</i>
6	Other	<i>The system should be easy to maintain, update, and deploy on Windows, Linux, and cloud platforms.</i>	<i>Easy deployment and future enhancement support</i>